

EREMU GRABITATORIOA

• INDAR GRABITATORIOA:

$$|\vec{F}| = G \frac{M_1 \cdot m_2}{d^2} \quad (N)$$

$$\vec{F} = m_2 \cdot \vec{g}$$

$$|\vec{g}| = G \frac{M_1}{d^2}$$

↳ $\left(\frac{N}{Kg} \text{ edo } \frac{m}{s^2} \right)$

• EREMUAREN INTENTSITATE
BEKTOREA EDO AZELERAZIOA

• EREMUKO PUNTU BATEKO
ENERGIA POTENTZIALA:

$$E_p = -G \frac{M_1 \cdot m_2}{d} \quad (J)$$

$$E_p = m_2 \cdot V$$

$$V = -G \frac{M_1}{d} \quad \left(\frac{J}{Kg} \right)$$

• EREMUKO PUNTU BATEKO
POTENTZIAL GRABITATORIOA:

• EREMUAK EGINDAKO LANA:

$$W_{\text{EREMUA}} = -\Delta E_p$$

• ABIADURA ORBITALA: ... $|\vec{F}_2| = |\vec{F}_G|$...

$$v_{\text{orb}} = \sqrt{G \frac{M_1}{d}} \quad \left(\frac{m}{s} \right)$$

• IHES ABIADURA: ... $E_{\text{m}} = E_{\infty}$...

$$v_i = \sqrt{2G \frac{M_1}{d}} \quad \left(\frac{m}{s} \right)$$

• ORBITAN EGONDA: E_m

$$E_m = \frac{1}{2} E_p$$

$$\text{eta } E_z = \frac{1}{2} |E_p|$$

• KEPLERREN 3. LEGEA: $\frac{T^2}{R^3} = K_G$

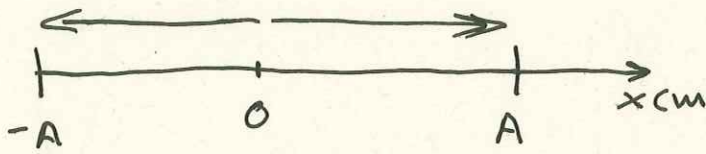
$$\frac{T_A^2}{R_A^3} = \frac{T_B^2}{R_B^3} = \dots = \frac{T_u^2}{R_u^3}$$

↳ ABIADURA ANGELUARRA $\omega = \frac{\text{angelua}}{\text{denbora}}$

$$\omega = \frac{2\pi}{T} \quad \left(\frac{\text{rad}}{s} \right)$$

eta $v = \omega \cdot R$

HHS (HIGIDURA HARMONIKO SINPLEA)



- ELONGAZIOA: $x(t) = A \sin(\omega t + \phi_0)$ (m)
- ABIADURA: $v(t) = \frac{dx(t)}{dt} = A\omega \cos(\omega t + \phi_0)$ (m/s)
- AZELERAZIOA: $a(t) = \frac{dv(t)}{dt} = -A\omega^2 \sin(\omega t + \phi_0)$ (m/s²)
 $\hookrightarrow$ edo $\hookleftarrow a(t) = -\omega^2 x(t)$

- PULTSATIOA: $\omega = \frac{2\pi}{T} = 2\pi f$ (rad/s) $T = \frac{1}{f}$
 PERIODOA: T (s) ; MAIZTASUNA EDO FREKUENTZIA: f (s⁻¹ edo Hz)

Hooke: $F = -k \cdot x$
 Nw. 2. L: $F = m \cdot a$ $a = -\omega^2 x$ $\Rightarrow K = m \cdot \omega^2 \rightarrow$

$\rightarrow$ OSZILADORE HARMONIKO SINPLEAREN PERIODOA $\rightarrow$

$\rightarrow K = m \cdot \frac{2^2 \pi^2}{T^2} \dots \rightarrow T = 2\pi \sqrt{\frac{m}{K}}$

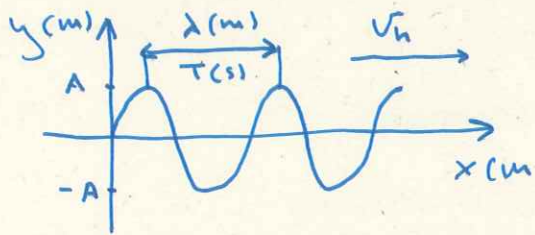
$\rightarrow$ Pendulo sinplean $T = 2\pi \sqrt{\frac{L}{g}}$ 

- ENERGIA MEKANIKOA (kte)

$E_m = E_p + E_z$ $E_p = \frac{1}{2} K x^2$ $E_z = \frac{1}{2} m v^2$

Muturrean $v=0$ eta $x=A$. Geldirik $\Rightarrow E_m = E_{pA} = \frac{1}{2} K A^2$

Uhin Harmoniko Sinpleak



Uhin Funtzioa

$$y(x,t) = A \sin(\omega t - kx + \phi_0) \quad (\text{m edo cm})$$

↑ Aurrera (Atzera +)

- PULTSAZIOA EDO MAIZTASUN ANGELARRA $\omega = \frac{2\pi}{T}$ edo $\omega = 2\pi \cdot f$ (rad/s)
- Uhin Kopurua EDO Uhin Zenbakia $k = \frac{2\pi}{\lambda}$ (rad/m)

• HEDAPEN ABIADURA: $v_h = \frac{\lambda}{T}$ (m/s)

• BIBRAZIO ABIADURA: $v(x,t) = \frac{dy(x,t)}{dt} = \dots = A\omega \cos(\dots)$

$v_{\max} \rightarrow \cos = \pm 1 \rightarrow v_{\max} = \pm A\omega$

• BIBRAZIO AZELERAZIOA: $a(x,t) = \frac{dv(x,t)}{dt} = -A\omega^2 \sin(\dots)$

$a_{\max} \rightarrow \sin = \pm 1 \rightarrow a_{\max} = \pm A\omega^2$

— 0 —
POTENTZIA $\Rightarrow P = \frac{E}{t}$ (wat = $\frac{J}{s}$)

INTENSITATEA = $\frac{\text{Potentzia}}{\text{azalera}} \rightarrow I = \frac{P}{S} \xrightarrow{\text{efektu}} I = \frac{P}{4\pi R^2}$ (wat/m²)

— 0 —

DESPASEAK

FASEA SIN BARRUKOA DA: $(\omega t - kx + \phi_0)$

KASUA:

→ $x_1 = 3\text{m}$ eta $x_2 = 5\text{m}$ -ko partikulen arteko desfasea:

$$\Delta\varphi_{3,5} = \varphi_3 - \varphi_5 = (\omega t - k \cdot 3 + \phi_0) - (\omega t - k \cdot 5 + \phi_0) = 2k \text{ (rad)}$$

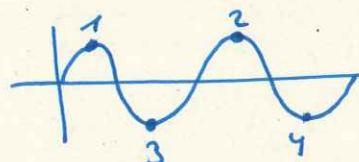
→ Zeinturien artean desfasea $\pi/3$ da?

$$\varphi_x - \varphi_{x+d} = \pi/3 \rightarrow (\omega t - kx + \phi_0) - [\omega t - k(x+d) + \phi_0] = \pi/3 \rightarrow \dots (d)$$

→ Zeinturik fasean $x = 2\text{m}$ -koaguz? $\Rightarrow$ FASEAN: $\Delta\varphi = n \cdot 2\pi$ ($n = \pm 1, \pm 2, \dots$)

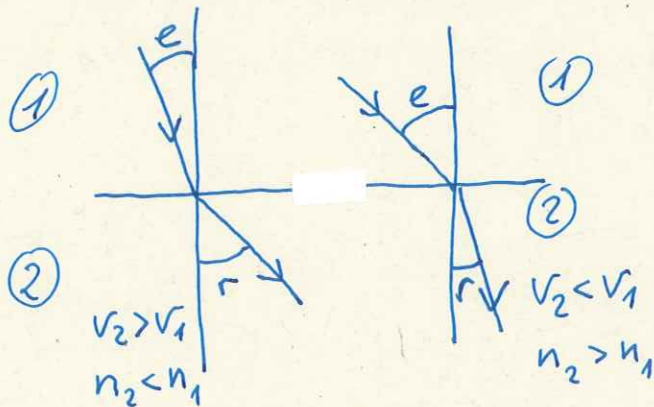
$$\varphi_2 - \varphi_{x_n} = n \cdot 2\pi \rightarrow (\omega t - k \cdot 2 + \phi_0) - [\omega t - k \cdot x_n + \phi_0] = n \cdot 2\pi \rightarrow x_n$$

→ Zeinturik kontrafasean? $\Rightarrow$ KONTRAFASEA $\Delta\varphi = (2n+1)\pi$ ($n = 0, \pm 1, \pm 2, \dots$)
(Garaipena Sardin)



1-2 $\Rightarrow$ FASEAN
1-3, 4 $\Rightarrow$ KONTRAFASEAN

ERREFRAKZIOA



f ez da aldatzen

$$v_i = \lambda_i \cdot f$$

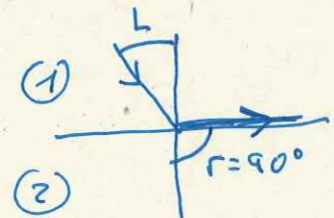
$$c = 3 \cdot 10^8 \text{ m/s}$$

- ERREFRAKZIO-INDIZEAK:
 - ABSOLUTUA $n_i = c/v_i$
 - ERLATIBOA $n_{21} = v_1/v_2$

• SNELL-EN ERREFRAKZIOAREN LEGEA

$$\frac{\sin e}{\sin r} = \frac{v_1}{v_2} = \frac{n_2}{n_1} = n_{21}$$

• ISLAPEN OSOA $\rightarrow$ BALDINTZA $v_2 > v_1$



$$\frac{\sin e}{\sin r} = \frac{v_1}{v_2} \xrightarrow[r=90^\circ]{e=L} \sin L = \frac{v_1}{v_2} \rightarrow L = \arcsin(v_1/v_2) = \arcsin(n_2/n_1)$$

Uhin ELEKTROMAGNETIKOAK



$$E(x,t) = E_0 \sin(\omega t - kx + \phi_0)$$

$$B(x,t) = B_0 \sin(\omega t - kx + \phi_0)$$

Eta beti: $\boxed{\frac{E}{B} = c}$

Uhin arruntak berala lantzen dira. Biak erlatibitateko edozein momentutan $\frac{E}{B} = c$ setehen da.

(Hau selektibitateko. Kurtsoan ez daf galdetzen)

EREMU ELEKTRIKOA

- INDAR ELEKTRIKOA:

$$|\vec{F}| = k \frac{|Q_1| \cdot |Q_2|}{d^2} \text{ (N)}$$

- EREMU ELEKTRIKOAREN

INTENSITATE BEKTOREA:

$$|\vec{E}| = k \frac{|Q_1|}{d^2} \left(\frac{\text{N}}{\text{C}} \right)$$

$$\vec{F} = \vec{E} \cdot Q_2$$

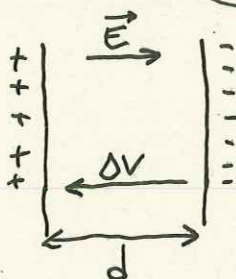
- ENERGIA POTENTIALA:

$$E_p = k \frac{Q_1 \cdot Q_2}{d} \text{ (J)}$$

$$E_p = V \cdot Q_2$$

- POTENTIAL ELEKTROSTATIKOA

$$V = k \frac{Q_1}{d} \left(\frac{\text{J}}{\text{C}} \text{ edo } \text{V} \right)$$



$$|\vec{E}| \cdot d = \Delta V$$

$\hookrightarrow$ ($\vec{E}$ bebai $\frac{\text{V}}{\text{m}}$)

- EREMU ELEKTRIKOAK EGINDAKO LANA:

$$W_{\text{EREM}} = -\Delta E_p$$

- ENERGIEN BALANTZAK:

- Eremua zentrala da, bera kontsektatzen, holan $E_m = k \cdot q \cdot \vec{r}$

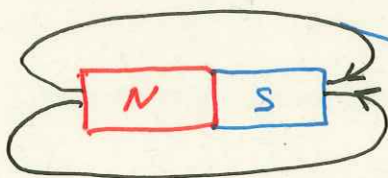
$$\rightarrow E_{mA} = E_{mB} \rightarrow E_{zA} + E_{pA} = E_{zB} + E_{pB} \rightarrow$$

$$\rightarrow \frac{1}{2} m_2 v_A^2 + Q_2 \cdot V_A = \frac{1}{2} m_2 v_B^2 + Q_2 \cdot V_B \rightarrow$$

$$\rightarrow \frac{1}{2} m_2 v_A^2 - \frac{1}{2} m_2 v_B^2 = Q_2 \cdot \Delta V$$

$\nwarrow$ Hortik joango gara, eta E_p kalkulatu kalkulatu (orkarrean)

EREMU MAGNETIKOA

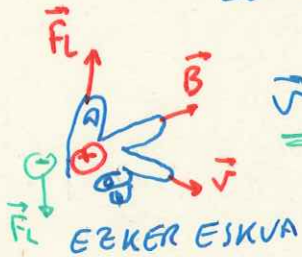


$\vec{B}$: indukzio bektorea (T, Tesla)

KARGA BATEN GAINEKO INDAR MAGNETIKOA

• LORENTZEN INDARRA:

$$\vec{F}_L = q \cdot (\vec{v} \wedge \vec{B}) = q \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ v_x & v_y & v_z \\ B_x & B_y & B_z \end{vmatrix}$$



$\vec{v} \perp \vec{B} \Rightarrow$ ibilbide zirkularra $\Rightarrow \vec{F}_L \equiv \vec{F}_z \rightarrow$

$$\rightarrow q v B \sin \alpha = m \frac{v^2}{R} \rightarrow \textcircled{R} \text{ lortu} \rightarrow \text{denpora} \rightarrow$$

$\uparrow_{90^\circ}$

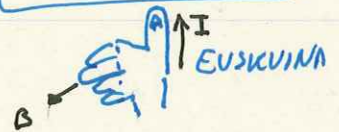
$$\rightarrow v = \frac{2\pi}{T} \cdot R \rightarrow \textcircled{T} \text{ lortu}$$

KORRONTE ELEMENTU BATEN GAINEKO INDAR MAGNETIKOA:

$$\vec{F}_L = q (\vec{v} \times \vec{B}) = q \left(\frac{\vec{\ell}}{\Delta t} \times \vec{B} \right) = \frac{q}{\Delta t} (\vec{\ell} \times \vec{B}) \Rightarrow \boxed{\vec{F}_L = I \cdot (\vec{\ell} \times \vec{B})}$$

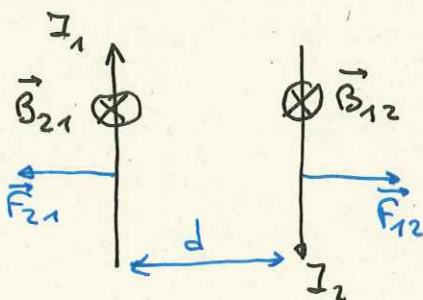
KORRONTE BATEK SORTURIKO EREMU MAGNETIKOA

BIOT-SAVART LEGEA



ESPIRA BATEN ERDIAN $B = \frac{\mu_0 I}{2R}$	KORRONTE ZUZEN INFINITUAK R DISTANTZIARA $B = \frac{\mu_0 I}{2\pi R}$
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BI KABLEEN ARTEKO ALKARREKINTZA

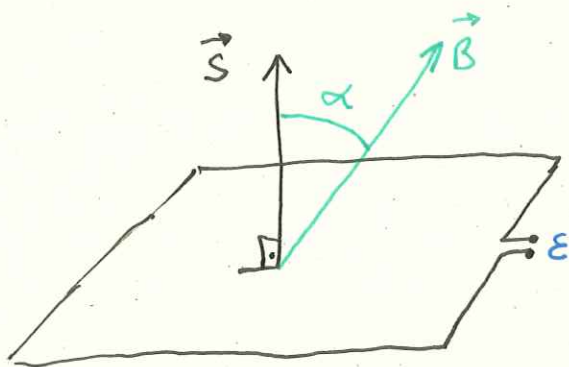


$$B_{12} = \frac{\mu_0 I_1}{2\pi d} \rightarrow F_{12} = I_2 \cdot l_2 \cdot B_{12} = \frac{I_2 l_2 \mu_0 I_1}{2\pi d}$$

$$B_{21} = \frac{\mu_0 I_2}{2\pi d} \rightarrow F_{21} = I_1 \cdot l_1 \cdot B_{21} = \frac{I_1 l_1 \mu_0 I_2}{2\pi d}$$

Edo l_1 eta l_2 eremoten deuskuve, edo bestela 1m jarri eta indarraren emaitza, adibidez: $8 \frac{N}{m}$

INDUKZIO ELEKTROMAGNETIKOA



• FLUXU MAGNETIKOA: ϕ

$$\phi = \vec{B} \cdot \vec{S} = B \cdot S \cdot \cos \alpha \quad (\text{Wb})$$

$S = \text{azalera}$

$\left[\begin{array}{l} \text{rectanguloa} \\ \text{zirkulua} \end{array} \right]$

$S = a \times b \quad (\text{m}^2)$
 $S = \pi R^2 \quad (\text{m}^2)$

• INDAR ELEKTROERAGILE INDUZITIBA: ε (V)

$\Rightarrow$ FARADAY-REN LEGEA

TARTEKA: $\varepsilon = - \frac{\Delta \phi}{\Delta t} \quad (\text{V})$

JARRAIA: $\varepsilon(t) = - \frac{d\phi(t)}{dt} \quad (\text{V})$

MINUSA
EZINBESTEKOA
LENZ-ERI JARRAITUZ

• KANPOKO ZIRKUITO BATERI ε -ek HORNITUTAKO KORRONTEA (I)
ZIRKUITUAN, R (Ω) ERRESISTENTZIA EGONDA $\Rightarrow$

$\Rightarrow$ OHM-EN LEGEA

$$I = \frac{\varepsilon}{R} \quad (\text{A})$$

$\vec{B}$: indukzio bektorea (T; Tesla)
 ϕ : fluxu magnetikoa (Wb; Weber)
 $\vec{S}$: azalera bektorea (m^2)
 ε : indar elektroeragile induzitiba (V; volt)
 R : erresistentzia (Ω ; ohm)
 I : intentsitatea (A; ampere)

OPTIKA GEOMETRIKO A - SARRERA + ISILUAK

• DIOPTRIO ESFERIKOEN OINARRIZKO EKUAZIOA

$$\frac{n'}{s'} - \frac{n}{s} = \frac{n' - n}{R} *$$

• DIOPTRIO BATEK POTENTZIA

$$P = \frac{1}{f'}$$

Unitatea: D (dioptrie)

• ALBO HANDIPENA: $A_L = \frac{y'}{y} = \frac{n \cdot s'}{n' \cdot s}$

$$f = \frac{n R}{n - n'} \quad f' = \frac{n' R}{n' - n}$$

• DIOPTRIO LAUA $\Rightarrow R = \infty \Rightarrow \frac{n'}{s'} - \frac{n}{s} = \frac{n' - n}{\infty} \Rightarrow \frac{n}{s} = \frac{n'}{s'}$

Albo handipena: $A_L = 1 \leftarrow \frac{n \cdot s'}{s \cdot n'} = 1 \leftarrow \dots$

• ISILU ETAN $\Rightarrow n' = -n$

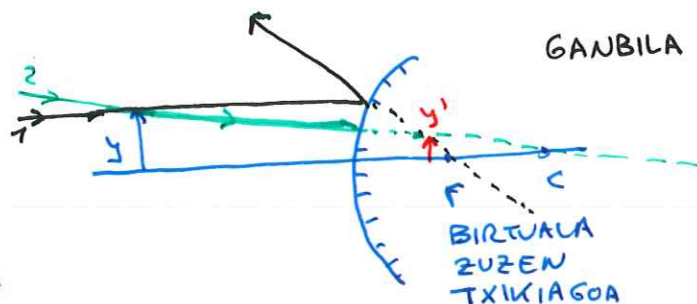
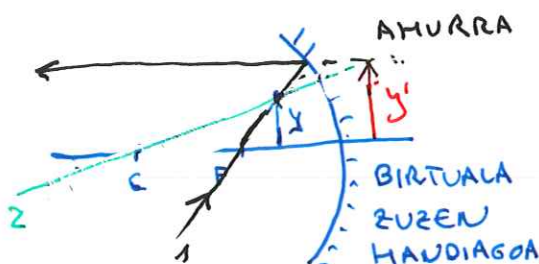
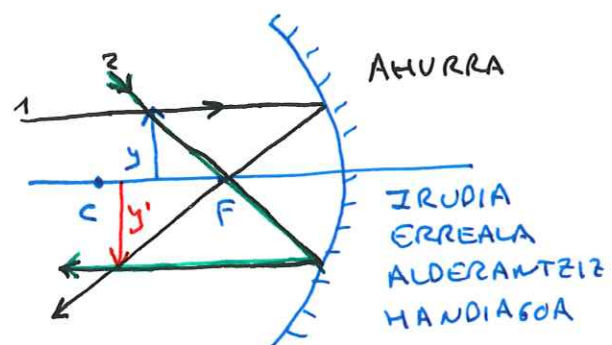
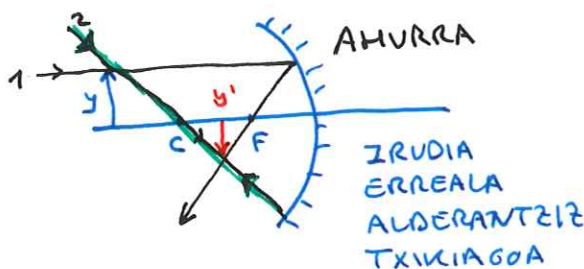
• ISILU LAUA $\Rightarrow R = \infty$ eta $n' = -n \Rightarrow$

$$\Rightarrow \frac{-n}{s'} - \frac{n}{s} = \frac{-n - n}{\infty} \Rightarrow \frac{-n}{s'} = \frac{n}{s} \Rightarrow \boxed{s' = -s} \quad \boxed{M_L = 1}$$

• ISILU ESFERIKOAK $\Rightarrow R$ eta $n' = -n$

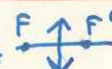
$$\Rightarrow \frac{-n}{s'} - \frac{n}{s} = \frac{-n - n}{R} \Rightarrow \dots \Rightarrow \boxed{\frac{1}{s'} + \frac{1}{s} = \frac{1}{f}} \quad \boxed{A_L = \frac{-s'}{s}}$$

Isilu esferikoetan $f = R/2$



- Irudi errealak ez dira ikusten, beharrik proiektzio pantaila batean
- Irudi birtualak ikusten ditugunak dira (isilu baten barruan adibidez)

OPTIKA GEOMETRIKOA - LEIARRAK

LEIAR MEHEAK (LENTE MEHEAK): KONBERGENTEAK  DIBERGENTEAK 

$$\frac{1}{s'} - \frac{1}{s} = \frac{1}{f} = -\frac{1}{f}$$

Emendioa:

$$\frac{y'}{y} = \frac{s'}{s}$$

Potentzia (dioptriak):

$$P = \frac{1}{|f|} \text{ dioptriak}$$

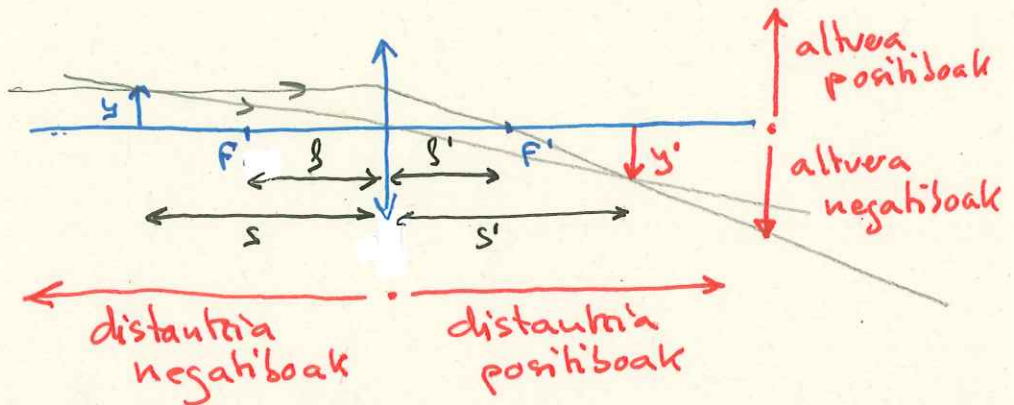
s: objektu-distantzia

s': irudi-distantzia

f: objektu distantzia fokala

f': irudi distantzia fokala

$$|f| = |f'| = \frac{R}{2} \quad (R: lentearen kurbadura erradioa)$$

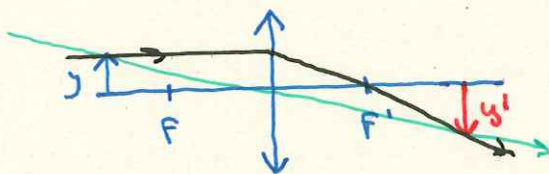


F: objektu fokua

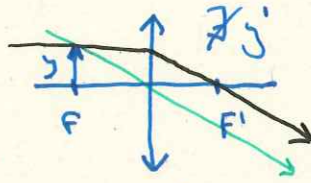
F': irudi fokua

y: objektuaren tamaina

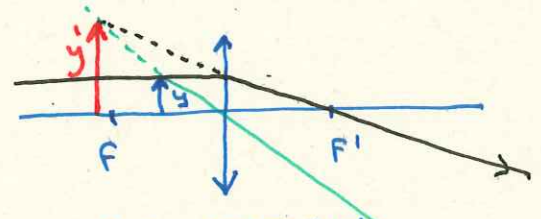
y': irudiaren tamaina



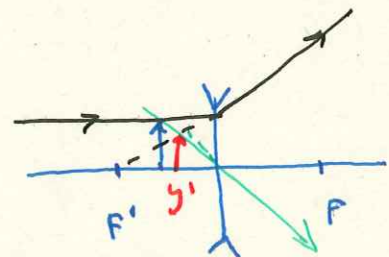
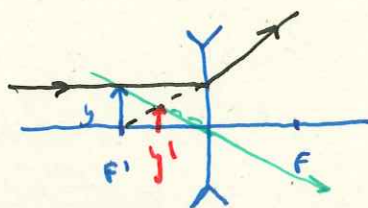
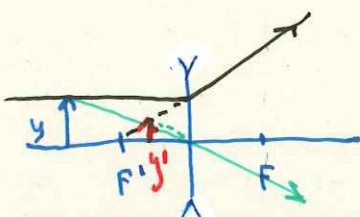
IRUDI ERREALA ALDERNTZIT



IRUDIRIK



IRUDI BIRTUALA ETA ZUZENA



DIBERGENTEETAN IRUDIA BETI: $\begin{cases} \text{BIRTUALA} \\ \text{ZUZENA} \\ \text{OBJEKTUA BAINO TXIKIAGOA} \end{cases}$

FISIKA MODERNOA

Fotoien energia $E = h \cdot f$ $h = 6.625 \cdot 10^{-34} \text{ J} \cdot \text{s}$

Potentzia: $P = \frac{E}{t}$ $\frac{\text{J}}{\text{s}} = \text{Watt}$

Intentsitatea: $I = \frac{W}{S}$ $\frac{\text{Watt}}{\text{m}^2}$

De Broglieen uhin-luzera: $\lambda = \frac{h}{p} = \frac{h}{m \cdot v}$

Efektu fotoelektrikoa: $W_{\text{max}} = h \cdot f - W_0$

Balaztaze potentziala $V_B \rightarrow q \cdot V_B = W_{\text{max}} \rightarrow V_B = \frac{1}{2} \frac{mv^2}{q}$

Atari maiztasuna = maiztasun minimoa fotoelektrikoki lotzeko.

$f < f_u \rightarrow \nexists$ fotoelektrik (edo $\lambda > \lambda_u$)

$f > f_u \rightarrow \exists$ fotoelektrik (edo $\lambda < \lambda_u$)